CALCULUS WITH

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TOPIC THREE --- ANSWER KEY | TOPIC SERIES

Derivatives

Compiled and taught by Asante

MATH 152 | Calculus & Analysis | KNUST BME1

Problem Set --- Worked Solutions

1. First principles: $f(x) = x^3$ **Correct answer: (A)**

$$f'(x) = \lim_{h\to 0} \frac{(x+h)^3 - x^3}{h} = \lim_{h\to 0} \frac{3x^2h + 3xh^2 + h^3}{h} = \lim_{h\to 0}(3x^2+3xh+h^2) = 3x$$

2. $f(x) = \cos(5x^2)$ **Correct answer: (B)**
   Chain Rule: outside derivative $-\sin(\cdot)$, inside derivative $10x$:

$$f'(x) = -\sin(5x^2)\cdot 10x = -10x\sin(5x^2)$$

3. $f(x) = x^3\cos x$ **Correct answer: (C)**
   Product Rule with $u = x^3$ ($u' = 3x^2$), $v = \cos x$ ($v' = -\sin x$):

$$f'(x) = 3x^2\cos x - x^3\sin x$$

4. $f(x) = \dfrac{x}{x^2+1}$ **Correct answer: (D)**
   Quotient Rule with $u = x$ ($u' = 1$), $v = x^2+1$ ($v' = 2x$):

$$f'(x) = \frac{(1)(x^2+1) - (x)(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$$

5. $f(x) = \sin^{-1}(2x)$ **Correct answer: (B)**
   Chain Rule with the inverse-sine derivative:

$$f'(x) = \frac{1}{\sqrt{1-(2x)^2}}\cdot 2 = \frac{2}{\sqrt{1-4x^2}}$$

6. Implicit: $x^3 + y^3 = 6xy$ **Correct answer: (A)**
   Differentiate both sides with respect to x (the right side needs the Product Rule too):

$$3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$$

   Collect $\frac{dy}{dx}$ terms:

$$\frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2 \implies \frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x}$$

7. Leibnitz: second derivative of $f(x) = x^3\ln x$ **Correct answer: (C)**
   With $u = x^3$ ($u' = 3x^2$, $u'' = 6x$) and $v = \ln x$ ($v' = 1/x$, $v'' = -1/x^2$):

$$(uv)'' = u''v+2u'v'+uv'' = 6x\ln x+2(3x^2)\left(\frac{1}{x}\right)+x^3\left(-\frac{1}{x^2}\right) = 6x\ln x+6x-x =$$

8. $\displaystyle\lim_{x\to 0}\frac{e^x-1}{x}$ **Correct answer: (D)**

9. $\lim_{x \to \infty} x \sin(1/x)$ **Correct answer: (B)**

   This is $\infty \cdot 0$. Rewrite as a quotient: $x \sin(1/x) = \dfrac{\sin(1/x)}{1/x}$. As $x \to \infty$, let $t = 1/x \to 0^+$, giving $\dfrac{\sin t}{t}$, which is the foundational trig limit equal to 1. So the limit is 1.

10. $\lim_{x \to 0^+} (1+x)^{1/x}$ **Correct answer: (D)**

    This is 1^∞. Let $y = (1+x)^{1/x}$, so $\ln y = \dfrac{\ln(1+x)}{x}$, which is $\frac{0}{0}$. By L'Hôpital's Rule:

    $$\lim_{x \to 0^+} \frac{\ln(1+x)}{x} = \lim_{x \to 0^+} \frac{1/(1+x)}{1} = 1$$

    So $\lim_{x \to 0^+} \ln y = 1$, meaning $\lim_{x \to 0^+} y = e^1 = e$ — this limit is exactly how e itself is often defined.

Cheat Sheet

The pages that follow are Paul Dawkins's widely used Calculus Cheat Sheet, included here as a fast-reference summary — not a replacement for working through this booklet. All credit for this reference sheet belongs to its original author.